

FINANCIAL MODELLING & ANALYSIS USING EXCEL

○ INTRODUCTION

Financial modelling is an important tool used in business to evaluate the financial performance of a project or company. It helps managers and investors understand how different financial factors such as cost, revenue, and investment affect profitability. Using tools like Microsoft Excel, financial models can be created to analyse various financial scenarios and support decision-making.

In this project, a financial model is developed using Excel to analyse the profitability of a hypothetical business. The model includes calculations such as revenue, total cost, profit, break-even point, payback period, and Weighted Average Cost of Capital (WACC). Additionally, sensitivity analysis and scenario analysis are performed to understand how changes in key variables impact the financial outcomes. Different charts such as break-even charts, waterfall charts, and bubble charts are also used to visually represent financial data.

The main objective of this project is to understand how financial modelling works and how Excel can be used to perform financial analysis for better business decisions.

○ ASSUMPTIONS

Dataset-

	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O
1	Fixed Cost	Variable Cost p	Selling Price per Unit	Project Investment	Expected Sales Volume	Unit Production Cap	Marketing Cost	Labor Cost	Maintenance Cost	Transportation Cost					
2	37564	97	117	328906	2502	2591	5582	8765	3843	2157					
3	18215	27	71	113245	1484	6469	9960	12851	1682	573					
4	34659	113	182	112136	1396	5768	9873	5197	3688	1761					
5	29831	127	178	532061	503	2289	5861	7920	4674	3389					
6	31473	56	166	902064	1331	4282	1135	13361	3338	2649					
7	44476	69	149	224521	1881	1908	4776	4719	1887	2374					
8	11522	188	323	688654	2374	3892	4272	6656	1674	1816					
9	30111	160	223	166563	3805	8278	4391	12711	3591	3000					
10	33902	177	283	628991	3177	5329	9630	14719	3591	827					
11	27972	67	179	430025	618	4946	3747	3363	1069	1288					
12	8001	115	166	249583	2704	9271	6645	11929	1867	2158					
13	12050	107	153	738383	3875	3699	2744	13417	3303	3918					
14	38998	100	183	353860	4845	2982	2766	13877	2444	2421					
15	13540	66	110	808393	1759	8540	2089	10701	738	541					
16	26141	193	253	801928	863	8050	2895	7305	1844	3065					
17	29665	144	230	454897	4075	1202	7593	9916	3882	1397					
18	11582	181	259	533821	748	8886	4957	11130	1453	3831					
19	25673	77	177	277983	2860	6712	4015	2797	2696	3386					
20	5500	105	176	702929	2348	7193	9941	8473	1767	3046					
21	5880	74	104	804288	1080	2881	5964	10695	3584	1005					
22	22668	26	170	153752	1571	9863	7348	12804	2635	1223					
23	14866	16	100	436077	812	4275	4581	13584	4451	1143					
24	22774	17	109	703249	4980	4150	6193	4177	2808	3755					
25	23169	86	142	742607	4998	6734	6367	11005	658	3910					
26	13005	172	256	177721	1187	3661	6208	3484	1733	3195					
27	21339	171	285	218441	4173	1898	2255	13119	2132	722					
28	44268	116	144	904921	1044	7738	2441	12398	674	3343					
29	33554	131	197	273669	2795	5244	2690	3293	1709	2334					

financial and operational dataset for a business/project. It contains 100 rows and 10 columns, where each row represents a different business scenario or product project with different costs, prices, and operational factors. It can be used for financial modeling, profit analysis, break-even analysis, and business decision making in tools like Excel.

Overall Structure

- Rows: 100 (each row = one project/product scenario)
- Columns: 10 (each column = a financial or operational variable)

Description of Each Column

1. Fixed Cost

This is the cost that does not change with production volume.

Examples:

- Rent
- Salaries of permanent staff
- Machinery setup cost

Even if the company produces 0 units or 1000 units, this cost remains the same.

2. Variable Cost per Unit

This is the cost required to produce one unit of product, and it changes with production quantity.

Examples:

- Raw materials
- Packaging
- Electricity used per unit

If production increases, total variable cost also increases.

3. Selling Price per Unit

This represents the price at which one unit of the product is sold to customers.

It is used to calculate:

- Revenue
- Profit
- Break-even point

Formula example:

Revenue = Selling Price × Units Sold

4. Project Investment

This is the initial amount of money required to start the project.

It may include:

- Equipment purchase
- Infrastructure
- Technology setup
- Initial operational funds

5. Expected Sales Volume

This column shows the estimated number of units the company expects to sell.

It helps estimate:

- Future revenue

- Profit
- Demand forecasting

6. Unit Production Capacity

This represents the maximum number of units that can be produced in a given time period (for example per month).

It helps determine whether production can meet expected demand.

7. Marketing Cost

This is the amount spent on advertising and promotion.

Examples:

- Social media ads
- Influencer marketing
- Posters and campaigns

Marketing costs help increase product awareness and sales.

8. Labor Cost

This refers to the wages paid to workers involved in production or operations.

It may include:

- Factory workers
- Technicians
- Temporary staff

9. Maintenance Cost

This is the cost required to maintain machines, equipment, or infrastructure so that operations run smoothly.

Examples:

- Machine servicing
- Repairs
- Replacement parts

10. Transportation Cost

This represents the cost of delivering products from the factory to warehouses or customers.

Examples:

- Fuel
- Logistics services
- Delivery vehicles

○ CALCULATIONS

- Revenue

➔ Revenue = Selling Price × Sales Volume

revenue
=C2*E2

revenue
5850
105364
254072
89534
220946
280269
766802
848515

- Total cost
- ➔ For calculating total cost we need find “total variable cost”.
Formula for that is, =Variable Cost × Sales Volume

Total cost = Fixed Cost + Total Variable Cost

total cost
=A2+Q2
total cost
5225
58283
192407
93712
106009
174265
457834
638911
596231
69378
318961
426675
523498

- Profit
- ➔ Profit = Revenue – Total Cost

profit
=P2-R2

profit
625
47081
61665
6555
114937
106004
308968
209604
302860
41244
129903

- Break-even unit

➔ Break-even Units = Fixed Cost / contribution per unit
Whereas contribution=(Selling Price – Variable Cost)

break even
=C2/\$2
break even
5.85
1.613636
2.637681
3.490196
1.509091
1.8625
2.392593
3.539683
2.669811
1.598214
3.254902
3.326087

- Payback Period

➔ Payback Period = Project Investment / Annual Profit

payback period	=B6/H18
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payback
7.309022
2.409468
2.156462
10.85839
17.02008
4.009304
11.87334
2.730541
10.48318
6.719141
3.725119
10.7012
4.983944
11.07388
10.69237
5.832013

- WACC (assume debt & equity structure)

➔ $WACC = (E/V \times Re) + (D/V \times Rd)$

Where

- **E** = Equity
- **D** = Debt
- **V = D + E**
- **Re** = Cost of Equity
- **Rd** = Cost of Debt

v	100000
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wacc $= (B8/B7 * B10) + (B7/H25 * B9)$

wacc
0.001033
0.00109
0.000991
0.001191
0.00109
0.000989
0.001247
0.001146
0.001046
0.00103
0.001188
0.001087
0.000987
0.001245

Part 2:

- Break-even analysis-

My assumptions

Descripti on	Value	Units Sold	Total Cost	Total Revenue
Fixed Cost	170000	500	320000	225000
Variable Cost per Unit	300	1000	470000	450000
Selling Price per Unit	450	1600	650000	720000
		2500	920000	1125000

- Scenario Analysis

My assumptions

Units	Selling Price	Variable Cost
200	510	300
250	530	280
100	430	320
150	500	210

Scenario Summary						
	Current Values:	Normal or Base	Best	Worst	Normal	Case study
Changing Cells:						
\$B\$3	590	590	600	450	590	590
\$B\$4	280	280	280	320	280	280
\$B\$5	250	250	250	150	250	250
Result Cells:						
\$B\$7	37500	37500	40000	-20500	37500	37500

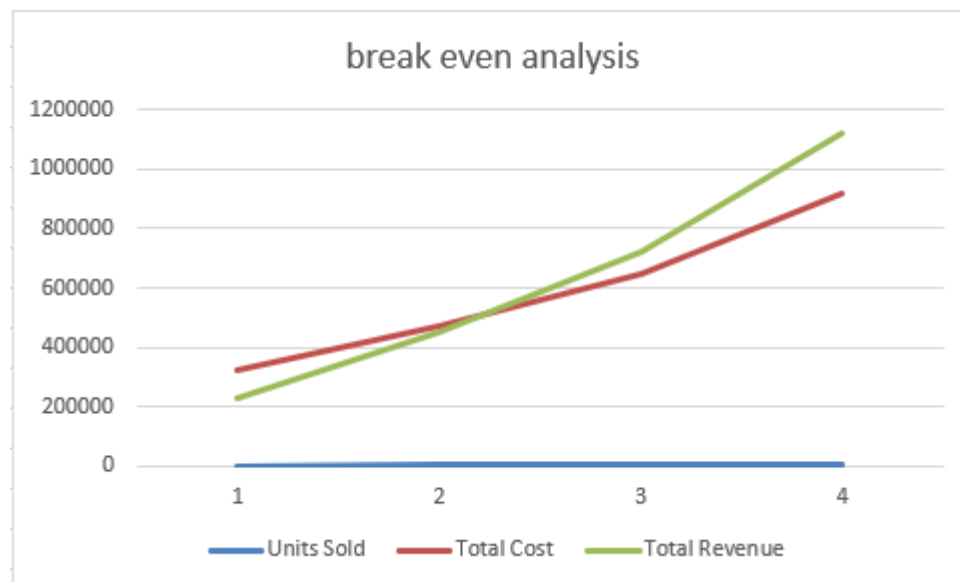
Interpretation

The analysis shows that the business performs well in normal and best conditions but may face a **large loss in the worst-case scenario**. Therefore,

managers should plan strategies to **reduce risks and maintain stable performance**.

Charts:

Break even-



Blue Line – Units Sold

- Shows the quantity of products sold at different levels.
- It remains close to the bottom because the scale is small compared to revenue and cost values.

Red Line – Total Cost

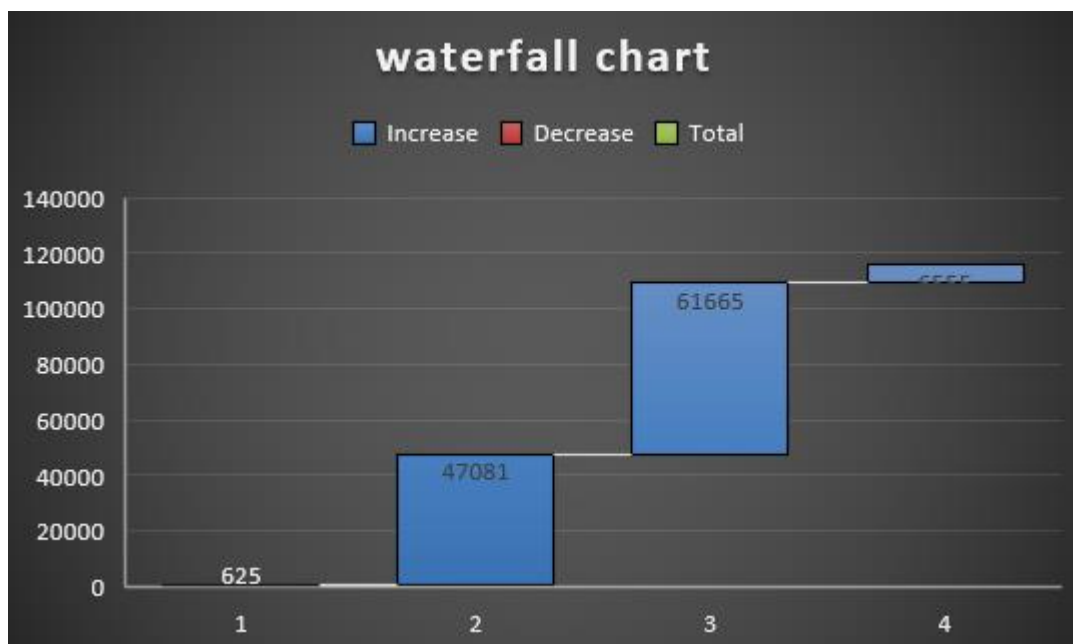
- Represents the **total cost of production**, including fixed and variable costs.
- The line increases as more units are produced because **variable costs increase with production**.

Green Line – Total Revenue

- Shows the **total income generated from selling products**.
- It increases more rapidly than total cost as sales increase.

This chart demonstrates how increasing sales affects costs and revenue. It shows that once the company sells **more than the break-even quantity**, it starts generating profit.

Water fall chart:



Interpretation-

The chart clearly shows that **most of the growth comes from the second and third stages**, while the first and fourth stages contribute smaller amounts. This helps analysts quickly identify **which factors have the biggest impact on the final value**.

Bubble chart:

(Sales volume vs marketing cost vs profit)



Interpretation-

Each bubble contains a number that represents the **marketing cost amount**:

- **5582** – Moderate marketing expenditure.
- **5861** – Slightly higher marketing spending.
- **1135** – Lowest marketing cost among the data points.
- **9820 and 9060** – Highest marketing costs, indicating major marketing investments.

The chart shows that **marketing spending varies significantly across campaigns**. Some campaigns require **high investment**, while others operate with **moderate or low budgets**. The visualization helps businesses **compare marketing efforts and identify where most resources are allocated**.

Dynamic chart: